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Polysaccharides from *Pleurotus ostreatus* alleviate cognitive impairment in a rat model of Alzheimer's disease

Yan Zhang[a,*], Xiaomei Yang[b], Gang Jin[a], Xiudong Yang[a], Yang Zhang[a]

[a] School of Chemical and Pharmaceutical Engineering, Jilin Institute of Chemical Technology, Jilin 132022, PR China
[b] Nutritional Department, Jilin Medical University Affiliated Hospital, Jilin 132013, PR China

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ABSTRACT

This study was conducted to evaluate the effects of polysaccharides extracted from *Pleurotus ostreatus* (POP) on D-galactose and $AlCl_3$-induced cognitive impairments. A behavioral test suggested that POP significantly decreased escape latency and increased crossing parameters of platform quadrant in a Morris water maze test. Furthermore, POP decreased error numbers and increased passive avoidance latency in a step-down test. Biochemical examinations revealed that POP significantly elevated superoxide dismutase, glutathione peroxidase, and catalase activities; and reduced malondialdehyde levels and acetylcholinesterase activity. Moreover, POP could decrease amyloid β peptide formation and tau phosphorylation by elevating the expression of protein phosphatase 2A as well as by reducing the expression of amyloid precursor protein (APP), β-site APP clearing enzyme1, and glycogen synthase kinase 3beta. These findings provide scientific evidence to support the exploitation of POP as a safe and effective drug to prevent and treat Alzheimer's disease.

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1. Introduction

Alzheimer's disease (AD) is the most common type of age-related dementia; it is characterized by hippocampal-dependent learning and memory deficits as well as behavior disorder [1]. Behavioral tests, including Morris water maze (MWM) test and passive avoidance step-down test, are always used to evaluate cognitive impairments in animal models. AD is also characterized by extracellular accumulation of insoluble deposits of amyloid β peptide (Aβ) and intracellular neurofibrillary tangles composed of aggregates of hyperphosphorylated tau protein. Sequential proteolytic cleavage of amyloid precursor protein (APP) by β site APP clearing enzyme1 (BACE1) is a crucial process in Aβ generation, which is considered to be a pivotal hallmark in the development and progress of AD [2]. A potential link also exists between the two pathological characteristics that Aβ accumulation causes an abnormal phosphorylation of tau by activating protein kinases such as glycogen synthase kinase 3beta (GSK3β) or by inhibiting phosphatases such as protein phosphatase 2A (PP2A), thereby resulting in neuronal death [3].

As of 2010, more than 36 million people worldwide suffered from AD. In spite of long-term research on the disease, no medicine has been developed as a cure for it. Although cholinesterase inhibitors and *N*-methyl-D-aspartate blockers, including tacrine, donepezil, rivastigmine, galantamine, and memantine, have been used as the main means to treat AD since 1993, these medications have limitations and even adverse effects such as stroke, vomiting, diarrhea, and insomnia [4]. Thus, the demand to focus on developing multifunctional natural plant as alternative therapeutic agents is increasing.

Pleurotus ostreatus is a widely cultivated edible mushroom that is a rich source of carbohydrates, proteins, vitamins, and minerals. *P. ostreatus* was belong to nontoxic materials according to the grade standards of mice acute toxicity. There were no death and organ pathology after maximally tolerating in 14 days in mice. The maximally tolerating dose (MTD) of *P. ostreatus* by oral administration was greater than 5000 mg/kg [5] and the MTD of polysaccharides from *P. ostreatus* was up to 4000 mg/kg [6]. *P. ostreatus* polysaccharides (POP) have a wide range of pharmacological effects such as anti-fungal [7], anti-oxidant [8,9], immunomodulatory [10], and anti-diabetic [11]. Until now, the anti-AD effect of POP on D-Galactose (D-gal) and $AlCl_3$(Al)-induced AD rats and the underlying mechanisms have not been investigated. Thus, in this study, the effect of POP on attenuating impairment of learning and memory was evaluated through behavioral tests and anti-oxidant effect of AD rats. Moreover, protein expressions of APP, BACE1, Aβ, GSK3β, PP2A, and phospho-tau (p-tau) were further measured by Western blot analysis to explore the underlying mechanisms.

* Corresponding author.
E-mail address: zyzxorange@126.com (Y. Zhang).

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2. Materials and methods

2.1. Chemicals and reagent kits

Fresh *F. ostreatus* was purchased from a local market (Jilin, China). D-Gal was purchased from Sigma-Aldrich (St. Louis, MO, USA). Al was purchased from Beijing Chemical Reagent Factory (Beijing, China). Total protein lysis buffer, BCA protein assay kit, and ECL reagents were acquired from Beyotime Institute of Biotechnology (Beijing, China). Antibodies against APP, Aβ1-42, p-tau (Ser235), GSK3β, BACE1, PP2A, and glyceraldehyde 3-phosphate dehydrogenase (GAPDH) were purchased from Proteintech (Rosemont, USA). Acetylcholinesterase (AchE) assay kit, superoxide dismutase (SOD) assay kit, catalase (CAT) assay kit, glutathione peroxidase (GSH-P_X) assay kit, malondialdehyde (MDA) assay kit, and total protein quantitative assay kit were obtained from Nanjing Jiancheng Bioengineering Institute (Nanjing, Jiangsu Province, China). All other reagents were of analytical grade and obtained from commercial sources.

2.2. Preparation of POP

Fresh *P. ostreatus* was dried at 50 °C, powdered, and sieved through a No. 300 mesh. The polysaccharides were prepared with distilled water using an ultrasonic-assisted cellulose method under optimized conditions (enzymatic hydrolysis temperature of 52 °C, enzymatic hydrolysis time of 57 min, enzymatic quantity of 0.0165 g, and ultrasonic time of 5 min). Crude polysaccharides extract was further deproteinized with Sevag reagent and dialyzed against deionized water, and then lyophilized as POP. The polysaccharide quality fraction of POP was 61% (w/w).

2.3. AD model construction and treatment

Male Wistar rats [Reg. No. SCXK (Jilin) 2015-0012] weighing 180 g to 220 g were purchased from the Experimental Animal Center, Jilin University (Jilin, China). The experiments were performed according to the guidelines established by the Guide for the Care and Use of Laboratory Animals of Jilin Institute of Chemical Technology, and all procedures were approved by the local Animal Ethics Committee.

Rats were divided into three groups: normal (15 rats), AD model (15 rats), and POP (15 rats). The AD model and POP groups were administered with Al (200 mg/kg, intragastrically) and D-gal (60 mg/kg, intraperitoneally) for 60 days, whereas the normal group was given vehicle. On the 30th day of modeling, the POP group began to be treated with POP (400 mg/kg) intragastrically. Meanwhile, the other two groups were administered with vehicle.

2.4. Behavioral test

2.4.1. MWM test

After 60 days of continuous feeding, all rats were placed in water to swim for 120 s towards a wall for acclimation. The place navigation and spatial probe tests were the two parts of MWM. Four quadrants of the maze pool (120 cm in diameter and 50 cm in height) were divided by the software and filled with water devitrified by milk to a depth of 17 cm, maintained at 24 ± 2 °C. A hidden platform with 10 cm in diameter was submerged 2 cm below the water surface and its position was maintained in the middle of the third (original) quadrant. Each rat was placed in the water from the same point to find the hidden platform during four trials each day for five consecutive days with a ceiling time of 120 s. If the rat failed to find the platform, it was gently guided toward the platform and placed on it for 20 s. The spatial probe test started from 24 h after the place navigation test was accomplished. Each rat was placed in the water from the quadrant opposite the original quadrant to find the removed platform. All the parameters of crossing the platform location and original quadrant were recorded and calculated by software and a video tracking system (Techman, China).

2.4.2. Step-down test

After the MWM test, a step-down test was conducted using a SDT-8 step-down platform recorder (Techman, China) to evaluate the memory function of the rats. Before the test, the rats were placed in reactive boxes for acclimation. The training and passive avoidance sessions were the two parts of the step-down test. In the training session, a 36 V foot shock was given to prompt the rats to look for the security platform in the corner. The number of times that the rats jumped down to the grid from the platform within 5 min was recorded as learning performances (first error numbers). The passive avoidance session was started 24 h after the training session. In the passive avoidance session, rats were first placed on the platform. The time period of rats staying on the platform was considered as the passive avoidance latency. Once the rats had jumped down and placed all four paws on the grid, the latency was ended. The numbers of rats that jumped down to the grid from the platform after latency within 5 min was recorded as memory performances (second error numbers). Any latency over 5 min was counted as 5 min.

2.5. Tissue and serum biochemical parameters

Both sides of the hippocampus and liver tissues were quickly removed, placed on ice, and homogenized in 9 vol. (w/v) of normal saline. The homogenate was 4 °C frozen centrifuged at 3500 rpm for 10 min and supernatant liquor was then obtained and stored at −80 °C. The whole blood of rats was centrifuged at 3000 rpm for 10 min at 4 °C to obtain the serum sample and then stored at −80 °C. Tissue homogenates and serum were used to evaluate AchE in the hippocampus as well as SOD, CAT, GSH-Px activities, and MDA level in the hippocampus, liver, and serum by using commercially available kits.

2.6. Hematoxylin and eosin (H–E) staining

The hippocampus samples were stored in 4% paraformaldehyde solution and then was stained with H–E to observe histopathological changes in the CA1 region of the hippocampus using standard light microscopy.

2.7. Western blot analysis

Western blot analysis was performed to detect protein expressions of APP, BACE 1, Aβ, GSK3β, PP2A, and p-tau in hippocampus tissues. Lysis buffer was used to pyrolyze the hippocampus tissues. Total protein was measured by a BCA assay kit using bovine serum albumin as standard. Proteins were electrophoretically separated by 5% sodium dodecyl sulphate polyacrylamide gel electrophoresis and then transferred to 0.45 μm polyvinylidene fluoride membranes (Millipore, USA). The membranes were immunoblotted with anti-APP, anti-Aβ, anti-BACE 1, anti-PP2A, anti-p-tau, anti-GSK3β, or anti-GAPDH antibodies. Visualization was performed using ECL reagents. Filter images were captured on a Gel Imaging System (Sim International Group, China). The relative protein levels were analyzed semiquantitatively using Bio-1D software (Sim International Group, China) and expressed in pixel units.

2.8. Statistical analysis

Normal data were expressed as mean ± S.E.M. Statistical comparison was performed using two-way analysis of variance on SPSS

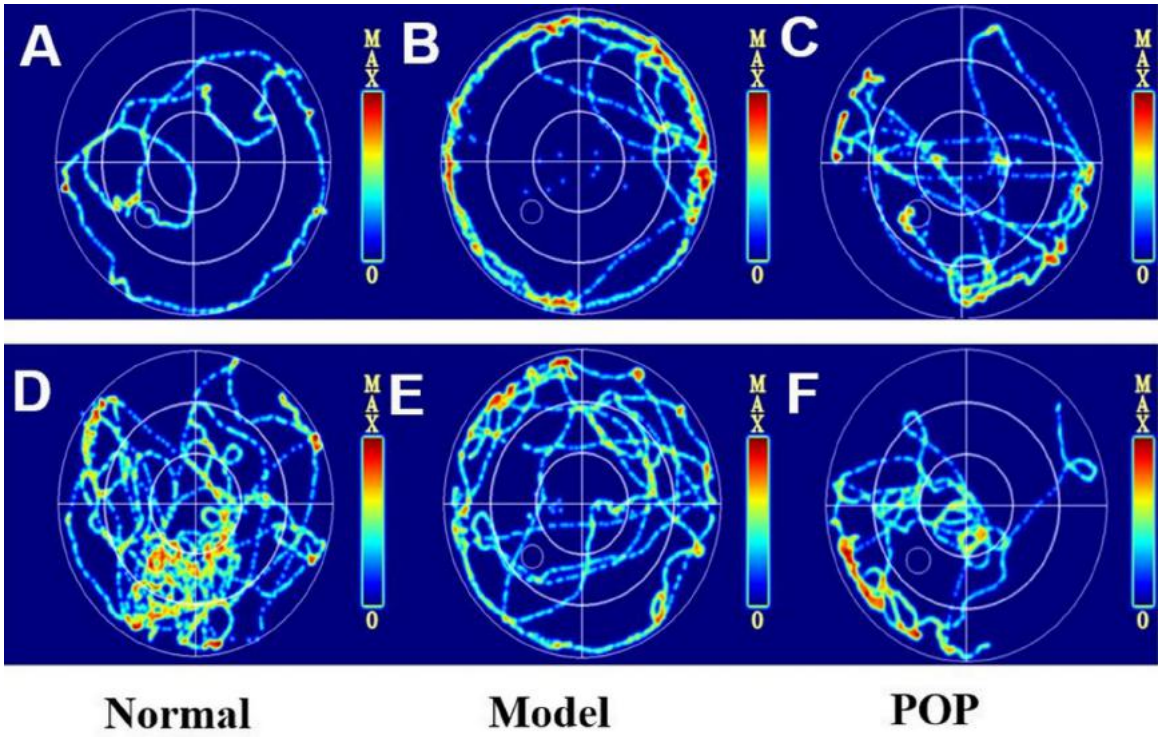

Fig. 1. The thermal infrared trajectory of Morris water maze test. (A–C) The trajectory of each group on the last day of place navigation test. (D–F) The trajectory of each group in probe test.

software, followed by Student's *t* test. p Values less than 0.05 were considered significant.

3. Results and discussion

3.1. Performance of behavioral test

Behavioral tests were always conducted to test cognitive impairments in AD models for the sensibility to hippocampal-dependent learning and memory deficits [12]. In the present study, swimming trajectories, escape latency, and crossing parameters were used as assessment criteria to evaluate the spatial learning and memory capability of rats by MWM test [13–15]. The error times and passive avoidance latency were used to assess memory acquisition and consolidation abilities by step-down test following a previously described method [12].

Fig. 1A–C showed the representative thermal infrared trajectories on the last day in the place navigation test of the three groups. The trajectory of the model rats was random swimming along the pool edge without purpose. However, the trajectories of the normal and POP rats were mostly swimming in the platform area and finally finding the hidden platform. Meanwhile, as shown in Fig. 2A–B, the escape latency to find the hidden platform for each day and the average latency for five days were shorter in the POP group than in the model group ($p < 0.05$ or $p < 0.01$). Fig. 1B–D showed the thermal infrared trajectories of the three groups in the spatial probe test. The model group continued to swim along the pool edge randomly without purpose. By contrast, the normal and POP groups swam in a straight direction and purposely crossed the platform quadrant to search for the removed platform. Fig. 2C–F showed the crossing parameters in the spatial probe test in which the crossing time, crossing distance, crossing times, and percentages in the platform quadrant of the POP group were significantly more than those of the model group ($P < 0.05$ or $P < 0.01$). All these results indicated that learning and memory impairment were attenuated by administration of POP. Fig. 3 showed the performance of the three groups in the step-down test. As shown in Fig. 3, the error numbers were fewer and the passive avoidance latency was longer in the POP group than in the model group, which demonstrated that the learning and memory performance of the AD rats significantly improved ($P < 0.05$ or $P < 0.01$) in the POP group.

3.2. Activity of AchE in hippocampus

Acetylcholine (Ach) is involved in several neuropsychiatric functions and plays a key role in learning and memory [16,17]. AchE is responsible for the degradation of Ach in the synaptic cleft. Inhibiting AchE activity can elevate Ach levels in the synaptic cleft and partially improve the cognitive impairment of AD patients [18].

As shown in Fig. 4, compared with the normal rats, the model rats exhibited an increase ($P < 0.01$) in hippocampal AchE activity. Compared with the model rats, the POP rats ($P < 0.05$) exhibited a decrease in hippocampal AchE activity, which implied that administering POP could lower the hippocampal AchE activity of AD rats.

3.3. SOD, GSH-Px, and CAT activities and MDA levels in hippocampus, liver, and blood

AD patients have been found to have a disorder of oxidant levels, which can be attributed to oxidant overproduction or antioxidant deficit. Evidence suggests that oxidative stress is connected with etiopathology of AD, including increasing Aβ-mediated neurotoxicity [19,20], causing mitochondrial dysfunction [21], enhancing tau hyperphosphorylation and neurofibrillary tangle formation [22], promoting neuron apoptosis and synaptic dysfunction [23]. Hydrogen peroxide (H_2O_2), superoxide (O^{2-}), hydroxyl anion (OH^-), hydroxyl radical (OH), nitric oxide (NO), and peroxynitrite (ONOO−) are considered as common oxidants that participate

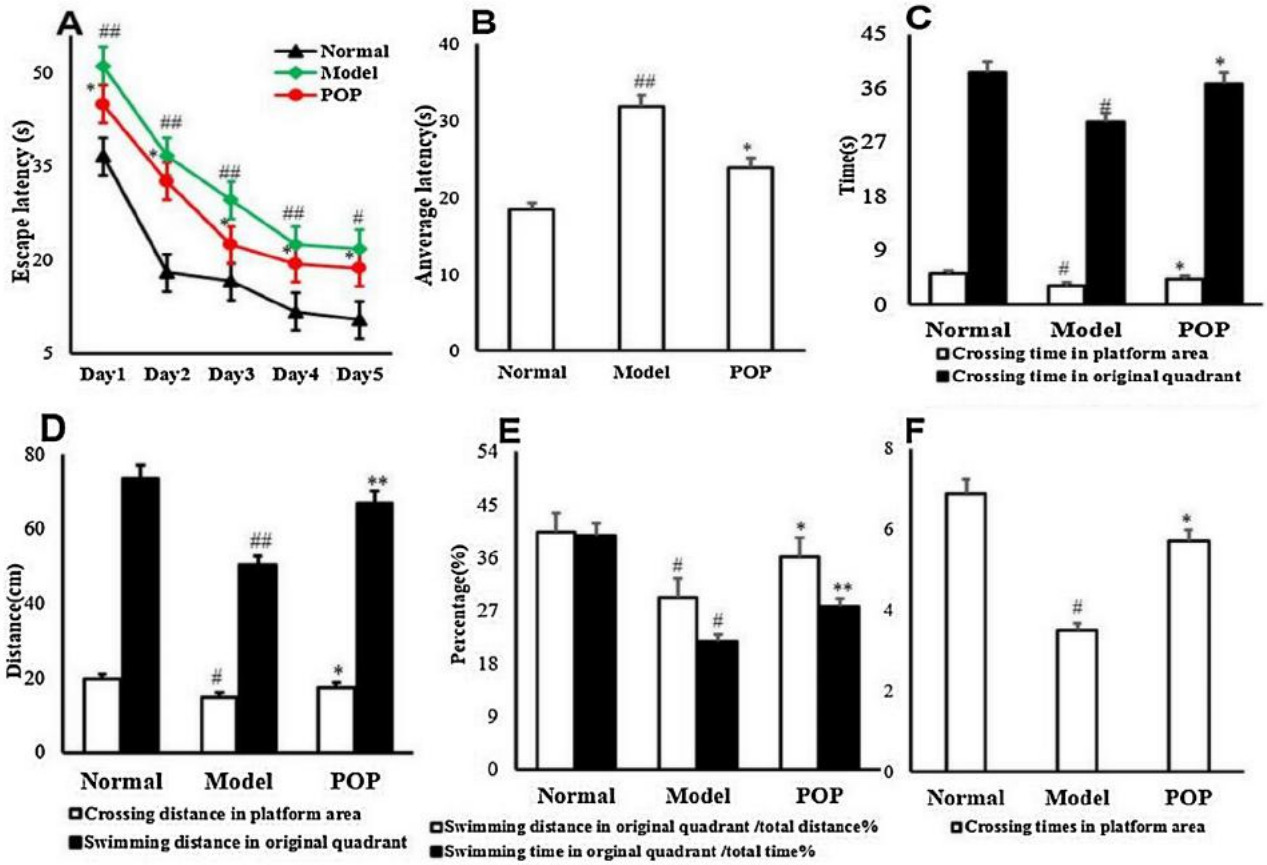

Fig. 2. The escape latency and cossing parameters of 3 groups in Morris water maze test. (A) The escape latency of five days in place navigation test. (B) The average escape latency of five days in spatial probe test. (C) The crossing time in platform area and original quadrant of spatial probe test. (D) The crossing distangce in platform area and original quadrant of spatial probe test. (E) The percentage of swimming distance and time of spatial probe test. (F) The crossing times in platform area of spatial probe test. #P < 0.05 versus normal group. ##P < 0.01 versus normal group. *P < 0.05 versus model group. **P < 0.01 versus model group.

in the redox state [24]. A strong correlation also exists between lipid peroxides, visualized as thiobarbituric acid-reactive species (TBARS), generally considered as MDA (the end product of lipid peroxidation) levels, antioxidant enzymes, and presence of senile plaques and neurofibrillary tangles in the AD brain [25]. GSH-Px is a primary means to deal with an excess of oxidants [26]. Mn-SOD and Cu/Zn-SOD are another means to eliminate oxidants by reducing superoxide to hydrogen peroxide [27]. CAT plays a part in removing H_2O_2, which is an enzyme that converts H_2O_2 to O_2 and H_2O [19,28].

As shown in Fig. 5, the MDA level increased and the SOD, GSH-Px, and CAT activities were significantly lower in the hippocampus, liver, and serum of model rats than those in the normal rats ($P < 0.01$ or $P < 0.05$). By contrast, the POP-administrated group significantly reversed these alterations. Compared with model rats, the POP rats exhibited a decreased MDA level while the SOD, GSH-Px, CAT and activities were significantly higher in the hippocampus, liver, and serum ($P < 0.01$ or $P < 0.05$). These findings indicated that POP could elevate antioxidant enzyme activities and alleviate oxidative stress in AD rats.

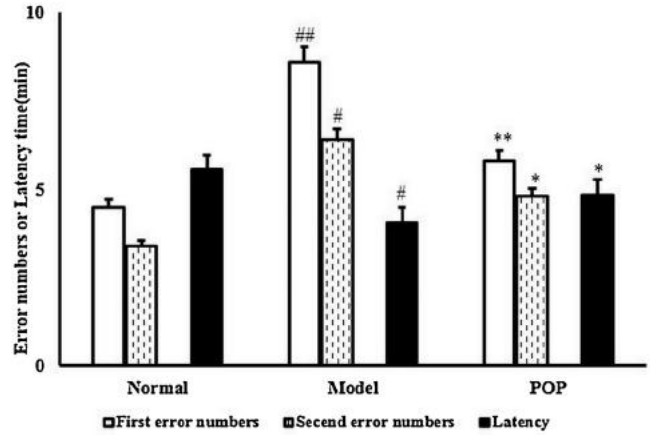

Fig. 3. The error times and passive avoidance latency of three groups in step-down test. #P < 0.05 versus normal group. ##P < 0.01 versus normal group. *P < 0.05 versus model group. **P < 0.01 versus model group.

3.4. Histological results

As shown in Fig. 6A, no noticeable hippocampal neuronal abnormalities were observed in the normal group. Fig. 6B shows the histological changes in model rats in which the CA1 pyramidal cells are arranged loosely and a remarkable loss of neurons occurs. The histological results of the model group suggested that modeling with D-gal combined with Al could lead to hippocampal impairment, an AD-like pathological feature. This pathological feature was changed in the group administrated with POP (Fig. 6C), which implied that POP could repair hippocampal impairment in AD rats.

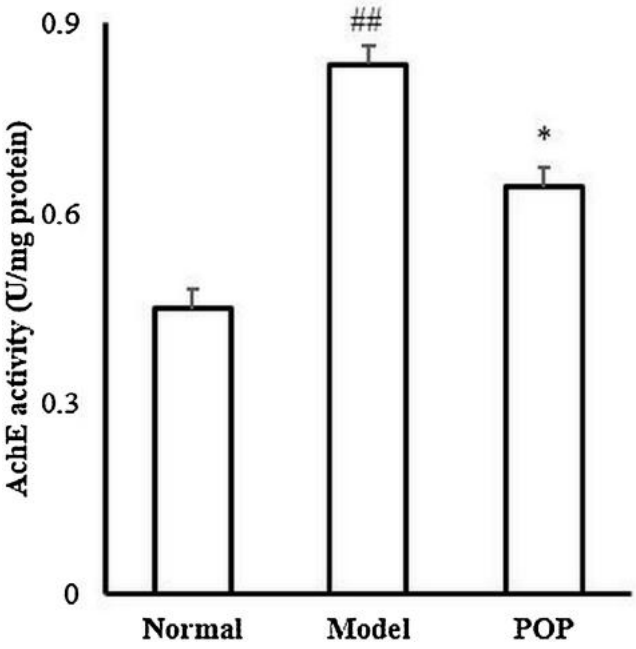

Fig. 4. Levels of AchE of three groups in rat hippocampus. ##P < 0.01 versus normal group. *P < 0.05 versus model group.

3.5. Protein expression in hippocampus

Aβ accumulation and tau protein hyperphosphorylation are two representative pathological characters of AD [29]. Aβ is produced by β-pathway proteolysis of APP [30]. When APP is cleaved by BACE 1, an N-terminal-soluble APP fragment (β-APPs) is released and a C-terminal APP fragment is cleaved by γ-secretase to create Aβ [3,31]. Tau phosphorylation level is mainly regulated by a balance between the action of various protein kinases and phosphatases [32]. Overexpression of protein kinases or inhibition of phosphatases to disrupt the balance could result in abnormal tau phosphorylation. For example, overexpression of tau kinases, such as GSK3β, in the brains of AD patients has been reported [33], and the phosphatase activity level of PP2A in the brains of AD patients has been reported to decrease by 50% [34]. A potential link also exists between the two pathological features of Aβ accumulation and tau hyperphosphorylation [35]. Aβ can result in abnormal tau phosphorylation by activating protein kinases, such as GSK-3β, and may cause neuronal death [36].

The Western blot results (Fig. 7) showed that the protein expressions of APP, Aβ, BACE 1, p-tau, and GSK3β increased and PP2A decreased in the hippocampus tissues of model rats compared with those of normal rats (P < 0.05 or P < 0.01). Meanwhile, compared with the model rats, the POP rats exhibited a reverse in the protein expressions of APP, Aβ, BACE1, p-tau; furthermore, GSK-3β decreased and PP2A increased (P < 0.05 or P < 0.01), which implied

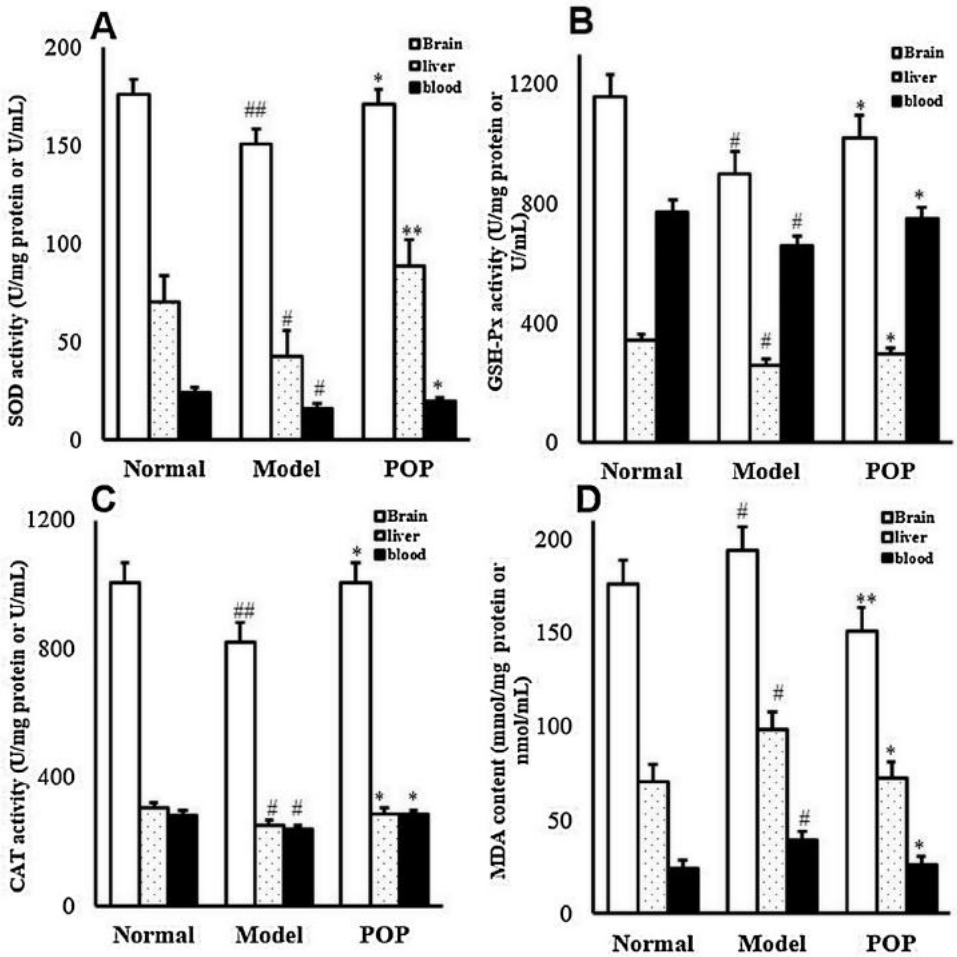

Fig. 5. The SOD(A), GSH-Px(B),CAT(C) activities and MDA(C) concentrations in hippocampus, liver and blood of AD rats. #P < 0.05 versus normal group. #P < 0.05 versus normal group. ##P < 0.01 versus normal group. *P < 0.05 versus model group. **P < 0.01 versus model group.

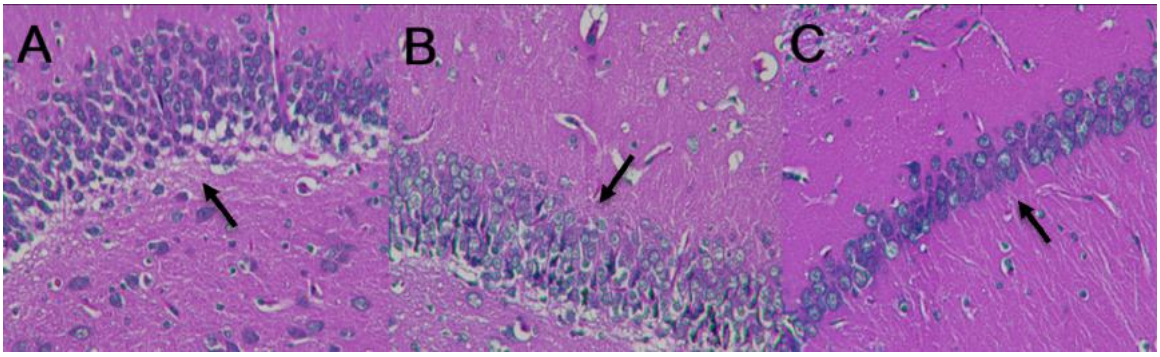

Fig. 6. H–E staining of pathologic changes in CA1 region of rats hippocampus. Solid arrow shows the histopathological changes in hippocampus. (Magnification: 400×, n = 4 in each group). Normal group(A), Model group(B), POP group(C).

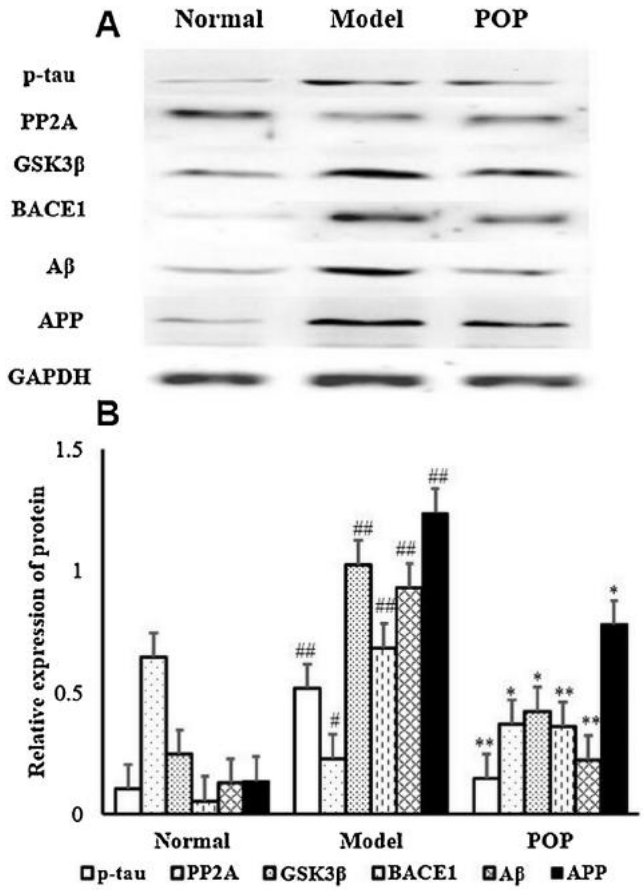

Fig. 7. Expression of p-tau, PP2A, GSK3β, BACE1 Aβ and APP in hippocampus of AD rats. (A) Western analysis of p-tau, PP2A, GSK3β, BACE1 Aβ and APP in hippocampus; (B) The relative protein density of p-tau, PP2A, GSK3β, BACE1 Aβ and APP in hippocampus. #P < 0.05 versus normal group. ##P < 0.01 versus normal group. *P < 0.05 versus model group. **P < 0.01 versus model group.

that the anti-AD effect of POP may be related to attenuating Aβ formation and tau phosphorylation.

4. Conclusion

When effectively administered, POP can alleviate cognitive impairment, decrease AchE activity, and attenuate oxidative stress. The possible mechanism of the beneficial effect is correlated with a decrease in Aβ formation and tau phosphorylation. All of these findings provide scientific evidence to support the exploitation of POP as a safe and effective drug to prevent and treat AD.

Acknowledgments

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References

[1] C. Supnet, I. Bezprozvanny, Cell Calcium 50 (2011) 303–309.
[2] K. Youn, J. Lee, E.-Y. Yun, C.-T. Ho, M.V. Karwe, W.-S. Jeong, M. Jun, J. Funct. Foods 10 (2014) 187–191.
[3] K. Yamada, T. Nabeshima, Pharmacol. Ther. 88 (2000) 93–113.
[4] H. Kavirajan, L.S. Schneider, Lancet Neurol. 6 (2007) 782–792.
[5] S. Vasudewa, D.T.U. Abeytunga, W.D. Ratnasooriya, Pharm. Biol. 45 (2007) 533–540.
[6] A. Ganeshpurkar, S. Kohli, G. Rai, Int. J. Med. Mushrooms 16 (2014) 207–217.
[7] J. Cilerdzic, M. Stajic, J. Vukojevic, I. Milovanovic, N. Muzgonja, Curr. Pharm. Biotechnol. 16 (2015) 179–186.
[8] G. Jaworska, K. Pogon, E. Bernas, A. Duda-Chodak, J. Food Qual. 38 (2015) 111–122.
[9] P.A. Fontes Vieira, D.C. Gontijo, B.C. Vieira, E.A.F. Fontes, L.S. de Assuncao, J.P.V. Leite, M.G.d.A. Oliveira, M.C.M. Kasuya, LWT-Food Sci. Technol. 54 (2013) 421–425.
[10] P. Maftoun, R. Malek, M. Abdel-Sadek, R. Aziz, H. El Enshasy, J. Sci. Ind. Res. 72 (2013) 655–662.
[11] Y. Zhang, T. Hu, H. Zhou, Y. Zhang, G. Jin, Y. Yang, Int. J. Biol. Macromol. 83 (2016) 126–132.
[12] M.H. Kim, S.H. Kim, W.M. Yang, Planta Med. 80 (2014) 1249–1258.
[13] Y. Wang, T.Z. Han, Neuroscience 160 (2009) 330–338.
[14] W. Yang, J. Yu, L. Zhao, N. Ma, Y. Fang, F. Pei, A.M. Mariga, Q. Hu, J. Funct. Foods 18 (2015) 411–422.
[15] M. Wen, L. Ding, L. Zhang, M. Zhou, J. Xu, J. Wang, Y. Wang, C. Xue, J. Funct. Foods 22 (2016) 417–430.
[16] S.B. Dunnett, B.J. Everitt, T.W. Robbins, Trends Neurosci. 14 (1991) 494–501.
[17] Z. Qu, J. Zhang, H. Yang, L. Huo, J. Gao, H. Chen, W. Gao, Physiol. Behav. 154 (2016) 114–125.
[18] P. Raina, P. Santaguida, A. Ismaila, C. Patterson, D. Cowan, M. Levine, L. Booker, M. Oremus, Ann. Intern. Med. 148 (2008) 379–W385.
[19] S. Miranda, C. Opazo, L.F. Larrondo, F.J. Munoz, F. Ruiz, F. Leighton, N.C. Inestrosa, Prog. Neurobiol. 62 (2000) 633–648.
[20] S. Jia, Z. Lu, Z. Gao, J. An, X. Wu, X. Li, X. Dai, Q. Zheng, Y. Sun, Int. J. Biol. Macromol. 83 (2016) 416–425.
[21] R. Yamaguchi, G. Perkins, Biochim. Biophys. Acta-Bioenerg. 1787 (2009) 963–972.
[22] P.K. Kamat, A. Kalani, S. Rai, S. Swarnkar, S. Tota, C. Nath, N. Tyagi, Mol. Neurobiol. 53 (2016) 648–661.
[23] M.P. Mattson, W.Z. Duan, J. Neurosci. Res. 58 (1999) 152–166.
[24] A.M. Swomley, D.A. Butterfield, Arch. Toxicol. 89 (2015) 1669–1680.
[25] M.A. Lovell, W.D. Ehmann, S.M. Butler, W.R. Markesbery, Neurology 45 (1995) 1594–1601.
[26] J. Blum, I. Fridovich, Arch. Biochem. Biophys. 240 (1985) 500–508.
[27] B. Halliwell, Nutr. Rev. 70 (2012) 257–265.
[28] M.Y. Aksenov, H.M. Tucker, P. Nair, M.V. Aksenova, D.A. Butterfield, S. Estus, W.R. Markesbery, J. Mol. Neurosci. 11 (1998) 151–164.
[29] C.-L. Lee, P.-Y. Lin, Y.-W. Hsu, T.-M. Pan, J. Funct. Foods 18 (2015) 387–399.
[30] J. Hardy, D. Allsop, Trends Pharmacol. Sci. 12 (1991) 383–388.

[31] J. Busciglio, D.H. Gabuzda, P. Matsudaira, B.A. Yankner, Proc. Natl. Acad. Sci. U. S. A. 90 (1993) 2092–2096.
[32] L. Martin, X. Latypova, F. Terro, Neurochem. Int. 58 (2011) 458–471.
[33] A. Hye, F. Kerr, N. Archer, C. Foy, M. Poppe, R. Brown, G. Hamilton, J. Powell, B. Anderton, S. Lovestone, Neurosci. Lett. 373 (2005) 1–4.
[34] F. Liu, I. Grundke-Iqbal, K. Iqbal, C.X. Gong, Eur. J. Neurosci. 22 (2005) 1942–1950.
[35] J. Busciglio, A. Lorenzo, J. Yeh, B.A. Yankner, Neuron 14 (1995) 879–888.
[36] A. Takashima, K. Noguchi, K. Sato, T. Hoshino, K. Imahori, Proc. Natl. Acad. Sci. U. S. A. 90 (1993) 7789–7793.